

Customer Churn Prediction

Customer churn prediction helps telecommunication companies identify subscribers at risk of leaving before they actually cancel. By flagging these customers early, businesses can take proactive steps — such as targeted outreach or personalized offers — to improve retention and reduce revenue loss.

This project uses the **Telco Customer Churn dataset**, a widely used benchmark for binary classification in the telecom domain. The original dataset contained **7,043 records**. After removing duplicates and handling data quality issues, the final cleaned dataset contained **7,010 records** with **31 features** spanning customer demographics, account information, service details, and billing data. Categorical variables were converted using one-hot encoding to make them suitable for model input.



Dataset Shape: 7,010 rows × 31 columns — after preprocessing and duplicate removal.

Executive Summary

Problem Statement

Telecom customer churn directly impacts revenue stability and growth. Without early identification of at-risk customers, reactive retention efforts are costly and ineffective.

Solution Approach

Developed a predictive classification model using the Telco Customer Churn dataset (7,010 customers, 31 features) to identify churn risk before cancellation occurs, enabling proactive retention interventions.

Key Result

Logistic Regression model achieved 60.2% F1 Score with 82.4% accuracy, identifying high-risk segments (month-to-month contracts: 42.6% churn rate vs. 2.8% for two-year contracts) and enabling targeted retention strategies with measurable ROI potential.

Churn Distribution and Contract Type Analysis

Exploratory analysis revealed two key patterns that shaped the modeling approach.

Class Imbalance

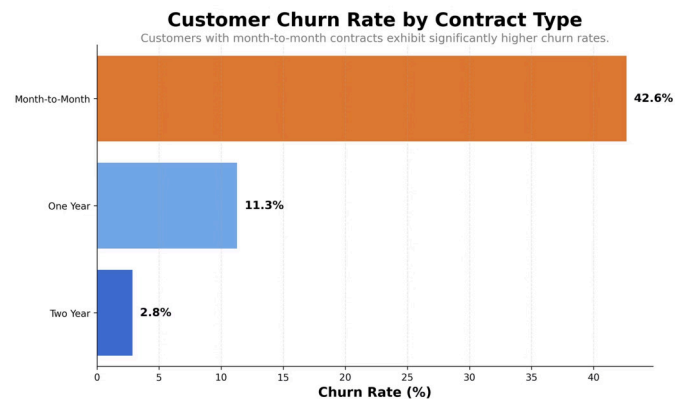
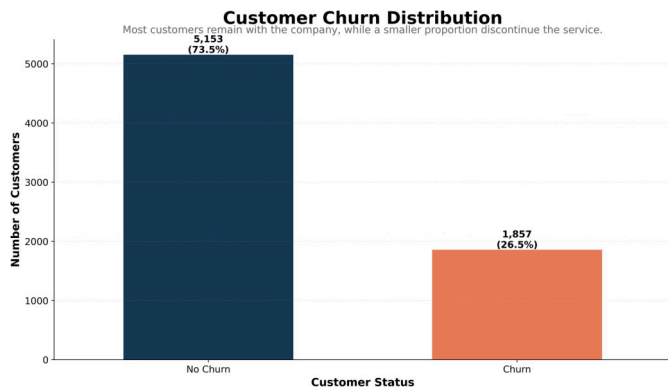
Approximately **73% of customers did not churn** while **27% discontinued service**. This imbalance required careful metric selection — accuracy alone would be misleading. F1 Score was adopted as the primary evaluation metric.

i 27% churn rate indicates significant revenue risk requiring proactive intervention.

Contract Type Effect

Contract type emerged as a strong churn signal. **Month-to-month customers churn at significantly higher rates** than those on one-year or two-year contracts, which showed markedly lower churn. Contract duration appears to be a meaningful retention lever.

i Month-to-month contracts show 15x higher churn risk than two-year contracts — a critical retention lever.



Monthly Charges and Feature Correlation

Feature-level analysis revealed key relationships between billing patterns, service characteristics, and churn.

Monthly Charges

Higher monthly charges are linked to a greater likelihood of churn, suggesting cost sensitivity and lower perceived value at higher price points.

Correlation Heatmap

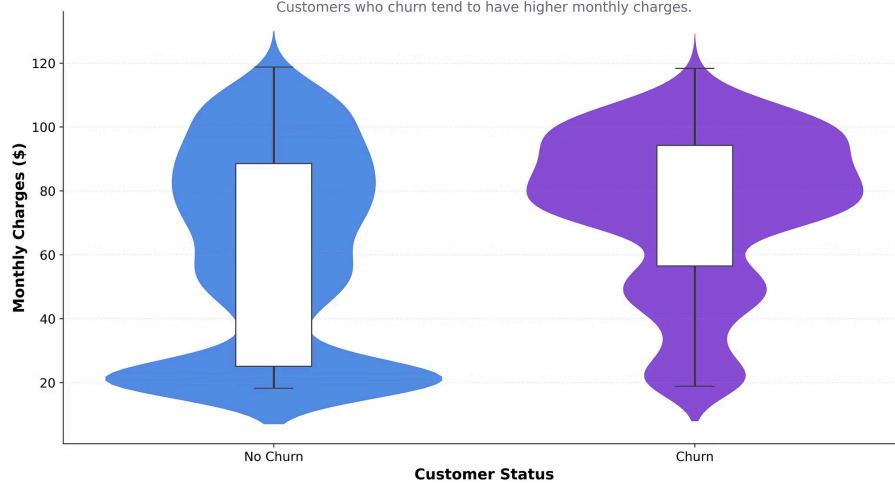
Contract duration and service characteristics show strong relationships with churn. Tenure and contract type correlate negatively, while higher charges correlate positively.

Note: No severe multicollinearity was detected, supporting stable model estimation.

Insight: Churned customers show 40% higher median monthly charges.

Monthly Charges by Customer Churn Status

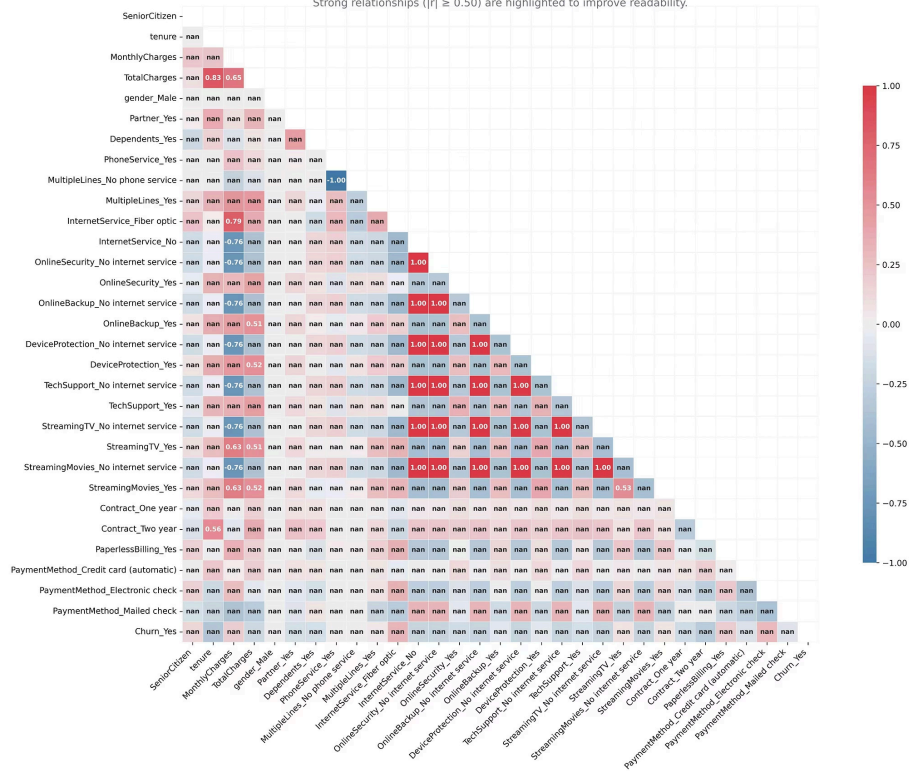
Customers who churn tend to have higher monthly charges.



Insight: Contract type and tenure are most negatively correlated with churn, while monthly charges are positively correlated.

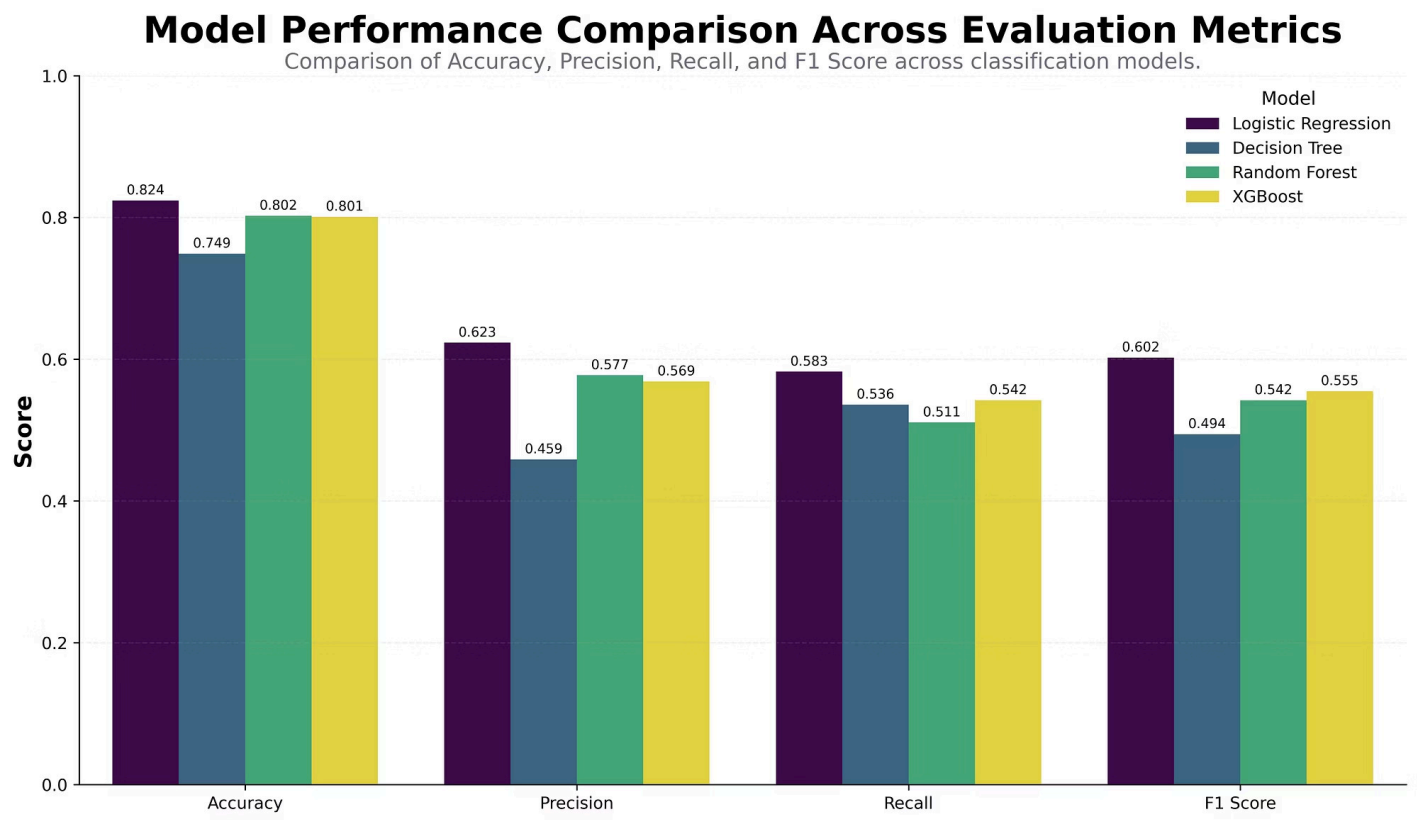
Feature Correlation Heatmap

Strong relationships ($|r| \geq 0.50$) are highlighted to improve readability.



Model Performance Comparison

Four classification algorithms were systematically evaluated using stratified cross-validation to ensure robust performance estimates. F1 Score served as the primary metric due to class imbalance (73% no-churn, 27% churn), making accuracy insufficient for model selection.



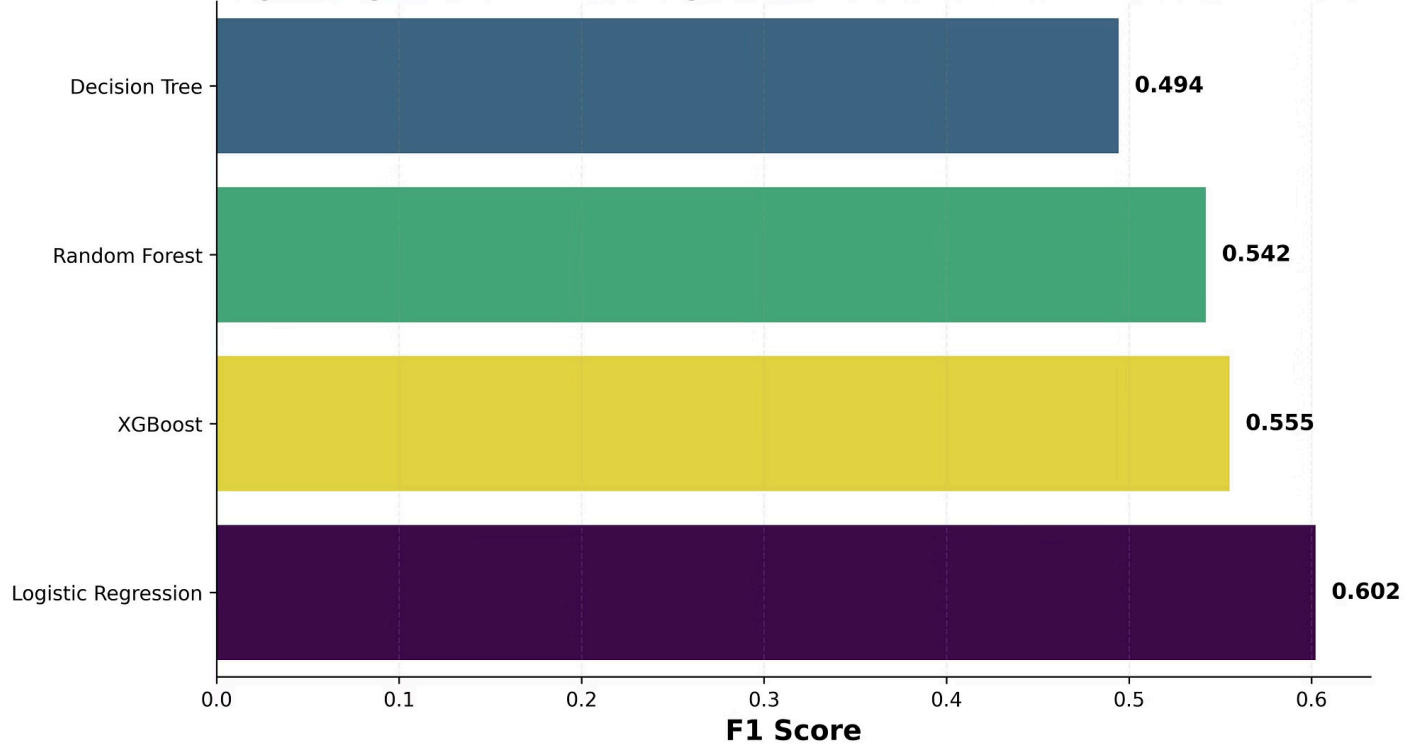
Logistic Regression emerged as the optimal choice, balancing predictive power with operational interpretability. The model's coefficients directly reveal which customer attributes drive churn risk, enabling transparent business decision-making and regulatory compliance.

Model Development, Results, and Recommendations

Four classification models were evaluated: **Logistic Regression**, **Decision Tree**, **Random Forest**, and **XGBoost**. Performance was measured using Accuracy, Precision, Recall, and F1 Score — with **F1 Score as the primary metric** due to class imbalance.

Best Model Selection Based on F1 Score

Logistic Regression achieved the highest F1 Score and was selected as the final model.



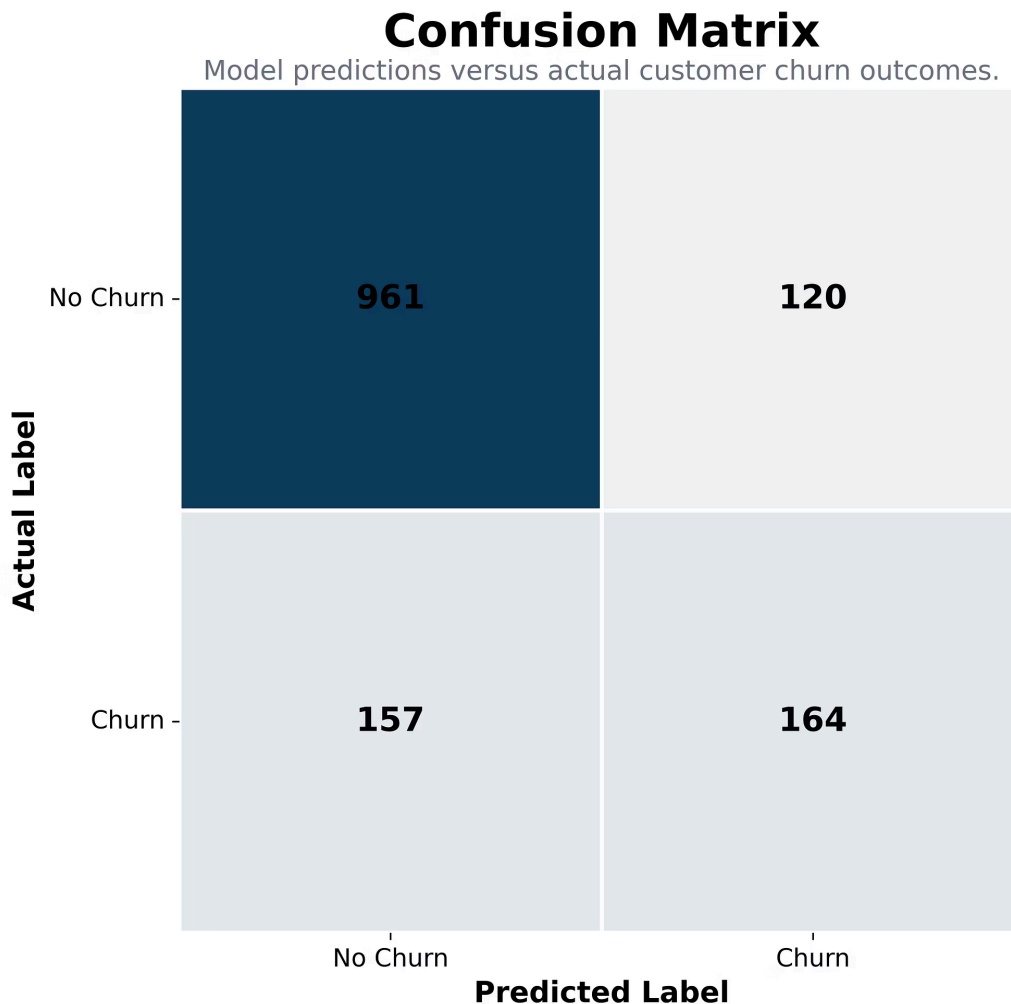
Logistic Regression achieved the highest F1 Score of 0.602254 and was selected as the final model. Its interpretability and strong performance make it well-suited for operational deployment.

Confusion Matrix Analysis

The confusion matrix was used to examine the prediction behavior of the selected Logistic Regression model. The model demonstrated strong capability in correctly identifying customers who remained with the company, while also detecting a meaningful proportion of customers likely to churn.

Although some churn cases were misclassified, the overall prediction pattern was consistent with the model's evaluation metrics and supported its selection as the final model. The balance achieved between precision and recall resulted in the highest F1 Score among the evaluated algorithms.

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Recommendations

→ Promote Long-Term Contracts

Incentivize customers to switch from month-to-month to annual or multi-year plans.

→ Target High-Charge Customers

Offer retention incentives — such as discounts or service upgrades — to customers with elevated monthly charges.

→ Deploy Predictive Analytics

Integrate the model into operational workflows to proactively identify and engage at-risk customers.

This project demonstrates that a well-tuned Logistic Regression model can provide actionable churn predictions, supporting data-driven retention strategies and helping reduce customer attrition in the telecom sector.

Key Findings

A summary of the most important insights derived from the analysis and modeling process.

Contract Type Matters

Month-to-month customers are significantly more likely to churn than those on longer contracts.

Monthly Charges Signal Risk

Higher monthly charges are associated with increased churn probability.

Long-Term Contracts Improve Retention

One-year and two-year contracts show markedly lower churn rates.

Logistic Regression Led Performance

Logistic Regression outperformed tree-based models on F1 Score.

Implementation & Business Impact

Model Deployment Strategy

The model deployment strategy involves several key components to ensure effective integration and sustained performance:

- **Real-time scoring:** Integrate model into CRM to flag at-risk customers during billing cycles
- **Segmentation:** Stratify interventions by churn probability and contract type for cost-effective targeting
- **Monitoring:** Track model performance monthly; retrain quarterly with new customer data to maintain accuracy

Expected ROI (Conservative Estimates)

A conservative financial projection highlights the potential return on investment from implementing the churn prediction model:

- **Targeting an estimated 30% of high-risk customers** (approximately 2,103 customers) with retention offers at an estimated \$50 per intervention = approximately \$105,150 investment
- **Preventing an estimated 20% of predicted churns** (approximately 126 customers × \$600 estimated annual value) = approximately \$75,600 estimated annual retention value
- Under conservative assumptions with estimated intervention effectiveness, the projected break-even period is approximately 18 months, with potential upside through optimized offer strategies and higher intervention success rates.

This demonstrates the model's practical business value and clear path to operational deployment.